Problem 7: Big Ben Forgor
10 Points

Problem ID: `forgor`
Rank: `3`

Introduction

After phishing the organizers of the contest by luring them in with the promise of a job interview and potential employment, Big Ben was able to steal a copy of the CALICO Fall '26 solutions for himself and see the sample test input for one of our problems. Unfortunately for him, the organizers were adept in the magic of Witch Hat Atelier, forcibly erasing all memories related to CALICO and transporting him to the isle of oblivion. Before his memory was wiped, however, Big Ben was able to write some numbers down on a piece of paper...

Problem Statement

This is a *run-twice (communication)* problem! Please read the Input/Output Format section *carefully* before attempting to solve the problem!

This problem has two phases:

1. In **Phase 1**, you will be given $\mathbf{N} - 1$ undirected edges $\mathbf{E_1}, \mathbf{E_2}, \ldots \mathbf{E_{N-1}}$ that compose a tree with $\mathbf{N}$ vertices, where the vertices are numbered 1 to $\mathbf{N}$. You must produce a list of $\mathbf{N}$ integers $\mathbf{A_1}, \mathbf{A_2}, \ldots, \mathbf{A_N}$. This list of integers will be stored by the checker. However, after outputting this list, your program will **terminate and all variables stored in memory will be lost**.
2. In **Phase 2**, you will be given $\mathbf{N}$ and the list of integers $\mathbf{A_1}, \mathbf{A_2}, \ldots, \mathbf{A_N}$ your program produced in Phase 1, but with one value replaced with -1. Your program should be able to reproduce and output the full original tree.

*Note: Given the unique input/output format, you may find the **Python, Java, and C++ templates** provided in the contest.zip useful.*

© 2026 CALICO Informatics Competition

Input/Output Format

The first line of the input contains a single string which is either `FIRST` or `SECOND` to denote the phase your program is currently being run in. Your program will be run twice per test set, once for each phase. The expected input and output differs based on the phase.

Phase 1

For this phase:

- The next line contains a single integer $\mathbf{N}$ denoting the number of vertices in the tree.
- Each of the next $\mathbf{N} - 1$ lines consists of two space-separated integers $\mathbf{U}_i$ $\mathbf{V}_i$ representing an edge in the tree.

For this phase, output a single line containing $\mathbf{N}$ space-separated integers $\mathbf{A}_1$ $\mathbf{A}_2$ ... $\mathbf{A}_\mathbf{N}$, representing the list that will be provided to you in **Phase 2**.

Phase 2

For this phase:

- The next line contains a single integer $\mathbf{N}$ denoting the number of vertices in the tree.
- The second line of the input consists of $\mathbf{N}$ space-separated integers $\mathbf{A}_1$ $\mathbf{A}_2$ ... $\mathbf{A}_\mathbf{N}$, which are identical to the integers produced by your program in **Phase 1**, but with one value replaced with -1.

For this phase, output $\mathbf{N} - 1$ lines consisting of two space-separated integers u_i v_i representing the edges in the original tree. The edges can be outputted in any order or orientation (i.e. u_i v_i and v_i u_i are considered identical for any u_i and v_i).

If the answer you output is correct, you will receive an accepted verdict. Otherwise, you will receive a wrong answer verdict. If at any point your program deviates from the interaction format (e.g. treats a **Phase 1** interaction as a **Phase 2** interaction), your program will be terminated. Your program will not be run any more and you will receive a wrong answer verdict.

Finally, we'd like to remind contestants that your program should only send and receive information through standard input/output as described above. Attempting to circumvent this by tampering with the judge system (for example by reading/writing files, modifying environment variables, etc.) is strictly forbidden and may result in disqualification. The judge will also attempt to use fake data/rollback attacks to catch suspicious submissions, but this will not count towards your time limit.

© 2026 CALICO Informatics Competition

Constraints

$2 \leq \mathbf{N} \leq 2 \cdot 10^5$
$0 \leq \mathbf{A}_i \leq 10^6$
Note the constraint on the values $\mathbf{A}_i$. If your program outputs values exceeding this constraint, your program will be terminated and you will receive a wrong answer verdict.

Sample Interaction

Sample Input Download

```
FIRST
3
1 2
1 3
```

Sample Output

```
1 2 5
```

Sample Input

```
SECOND
3
-1 2 5
```

Sample Output

```
3 1
1 2
```

Note that this is one of many possible correct outputs. If there are multiple solutions, you may output any of them.

Sample Explanations

For the test case, our program first reads `FIRST` for **Phase 1** and an input tree with 2 edges. Based on its implementation, it chooses to output `1 2 5`. Note that this does not necessarily represent an effective or optimal scheme.

The program is then terminated and rerun for **Phase 2**, where it reads `SECOND` and its own output from **Phase 1**. In this execution, $\mathbf{A}_1$ is replaced with -1. The program outputs edges `3 1` and `1 2`, which is accepted as correct. Note that any ordering and orientation of the output does not affect correctness.

© 2026 CALICO Informatics Competition